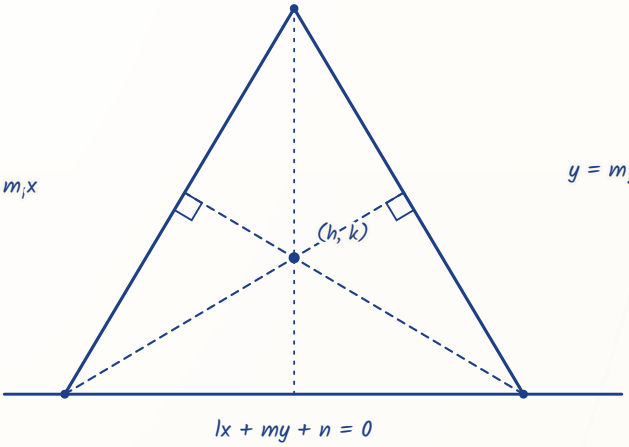
If $ax^2 + 2hxy + by^2 = 0$ rep two sides of triangle and $lx + my + n = 0$ rep third side of triangle.

$(y - m_1x)(y - m_2x) \rightarrow ax^2 + 2hxy + by^2 = 0$

$y = m_1x$ and $y = m_2x$

$m_1 \cdot m = -1$

$m = \dfrac{-1}{m_1}$



$\dfrac{-1}{m_2}$

$\dfrac{-1}{m_2} \cdot m_2 = -1$

$y = m_1x \;,\; lx + my + n = 0$

$lx + m\,m_1\,x + n = 0$

$x\,(l + mm_1) + n = 0$

$x = \dfrac{-n}{l + mm_1}$ similarly for $y = m_2x$

$x = \dfrac{-n}{l + mm_2}$

$y = m_1x \rightarrow \dfrac{y}{m_1} = x$

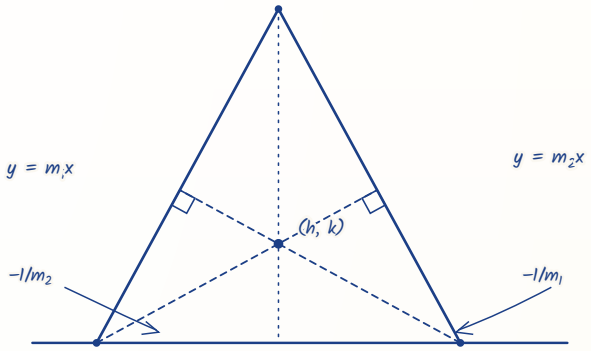
$lx + my + n = 0$

$l \cdot \dfrac{y}{m_1} + m\,y + n = 0$

$ly + m\,m_1\,y + m_1\,n = 0$

$y\,(l + mm_1) = -\,m_1\,n$

$y = \dfrac{-n\,m_1}{l + mm_1}$ similarly $y = \dfrac{-n\,m_2}{l + mm_2}$



$\frac{-n}{1+mm_1}, \frac{-n\,m_1}{1+mm_1}$ $\qquad$ $\frac{-n}{1+mm_2}, \frac{-n\,m_2}{1+mm_2}$

$$\frac{y-k}{x-h} = -\frac{1}{m_2}$$

$$\frac{\frac{-n\,m_1}{1+mm_1} - k}{\frac{-n}{1+mm_1} - h} = -\frac{1}{m_2}$$

$$\frac{\frac{-n\,m_1 - k(1+mm_1)}{1+mm_1}}{\frac{-n - h(1+mm_1)}{1+mm_1}} = -\frac{1}{m_2}$$

$$m_2\,[-n\,m_1 - k(1+mm_1)] = -[-n - h(1+mm_1)]$$

$$-n\,m_1\,m_2 - k\,m_2\,(1+mm_1) = n + h(1+mm_1)$$

$$-n\,m_1\,m_2 - n = (h + k\,m_2)(1+mm_1)$$

$$-n\,(m_1\,m_2 + 1) = (h + k\,m_2)(1+mm_1)$$

$ax^2 + 2hxy + by^2 = 0$

$$\frac{a\,x^2}{x^2} + \frac{2hxy}{x^2} + \frac{b\,y^2}{x^2} = 0$$

$$a + 2h\frac{y}{x} + b\left(\frac{y}{x}\right)^2 = 0$$

$c/a \Rightarrow$ product of roots

$m_1\,m_2 = a/b$

$m_1 + m_2 = -2h/b$

① $-n\left(\frac{a}{b}+1\right) = (h + k m_2)(1 + m m_1)$

similarly on equating

$$\frac{\frac{-n m_2}{1 + m m_2} - k}{\frac{-n}{1 + m m_2} - h} = -\frac{1}{m_1}$$

on simplifying

$m_1 [-n m_2 - k (1 + m m_2)] = - [-n - h (1 + m m_2)]$

$-n m_2 m_1 - k m_1 (1 + m m_2) = + n + h (1 + m m_2)$

$-n\left(\frac{a}{b}+1\right) = h (1 + m m_2) + k m_1 (1 + m m_2)$

② $-n (a/b + 1) = (h + k m_1)(1 + m m_2)$

① & ②

$$(h + k m_1)(1 + m m_2) = (h + k m_2)(1 + m m_1)$$

~~hl~~ $+ m h m_2 + k l m_1 +$ ~~$k m m_1 m_2$~~ $=$ ~~hl~~ $+ h m m_1 + k l m_2 +$ ~~$k m m_1 m_2$~~

$k l m_1 + m m_2 h = h m m_1 + k l m_2$

$k l (m_1 - m_2) = h m (m_1 - m_2)$

~~$kl \left(\left(\frac{2h}{b}\right)^2 - \frac{4a}{b}\right) = hm (-2h)$~~

$$kl = hm$$

equating to constant

$$kl = hm$$

$$\frac{k}{m} = \frac{h}{l} = t$$

$$k = mt \quad | \quad h = lt$$

sub both in eqn ①

~~$t\,(l + m\,m_2)(l + m\,m_1) = -n\,(m_1\,m_2 + l)$~~

$$t = \frac{-n\,(m_1\,m_2 + l)}{(l + m_1\,m)(l + m\,m_2)}$$

~~$t = n$~~

$$-n\,\frac{a+b}{b} = (l\,t + m\,t\,m_2)(l + m\,m_1)$$

$$-n\,\frac{a+b}{b} = t\,(l + m\,m_2)(l + m\,m_1)$$

$$t = \frac{-n\,(a+b)/b}{l + m\,m_2 + m^2\,(a/b)}$$

$$-n\,\frac{a+b}{b} = l^2\,t + l\,t\,m\,m_1 + m\,m_2\,l\,t + m^2\,m_1\,m_2\,t$$

$$-n\,\frac{a+b}{b} = t\,\left(l^2 + l\,m\,m_1 + m\,m_2\,l + m^2\,\frac{a}{b}\right)$$

$$-n\,\frac{a+b}{b} = t\,\left(l^2 + l\,m\,(m_1 + m_2) + \frac{a\,m^2}{b}\right)$$

$$-n\,\frac{a+b}{b} = t\,\left(l^2 + l\,m\,\frac{-2h}{b} + \frac{a\,m^2}{b}\right)$$

$$-n\,\frac{a+b}{b} = t\cdot\frac{b\,l^2 - 2\,h\,l\,m + a\,m^2}{b}$$

$$t = \frac{-n\,(a+b)}{b\,l^2 - 2\,h\,l\,m + a\,m^2}$$

$$h \Rightarrow h = l\,t \Rightarrow t \times l = \frac{-n\,(a+b)}{b\,l^2 - 2\,h\,l\,m + a\,m^2} \times l = \frac{-n\,l\,(a+b)}{a\,m^2 - 2\,h\,l\,m + b\,l^2}$$

$$k \Rightarrow k = m\,t \Rightarrow t \times m = \frac{-n\,(a+b) \times m}{a\,m^2 - 2\,h\,l\,m + b\,l^2} = \frac{-n\,m\,(a+b)}{a\,m^2 - 2\,h\,l\,m + b\,l^2}$$

orthocentre (h, k)

$$h = \frac{-n\,l\,(a+b)}{a\,m^2 - 2\,h\,l\,m + b\,l^2}$$

$$k = \frac{-n\,m\,(a+b)}{a\,m^2 - 2\,h\,l\,m + b\,l^2}$$